Complexity Is Nature's Way of Exploiting Energy Effectively

An order of construction, its ceiling, and why two

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2026-09-05

Abstract

We treat the emergence of complexity as a ledger problem: a form has a build cost, a maintenance cost, and a function that must repay both with a surplus. Working in the retention framework of the companion paper, where a record is the entropy of the classes a dynamics cannot mix and H = R + H_thermo, we obtain an ORDER OF CONSTRUCTION and a CEILING, both from the accounting rather than from biology. Nothing begins unless some function costs nothing to build (2000 of 2000 parameter sets stall at zero rungs otherwise), which forces the first tier to be *geometry* – a cup that forms, a hole with a size, a flow that sorts by speed – because geometry holds no record and so costs nothing to maintain. Digestion necessarily precedes selection (first in 100% of 2000 sweeps that started; selection first in 0 of 2000), because a selective return is bounded by a throughput that does not yet exist. The ladder is driven by surplus, not capability: a form that exactly repays its own maintenance is adoptable and contributes nothing, and with only such forms the ladder stalls at the first rung. Complexity then compounds in bursts – waits shorten within a form type, lengthen as geometric build costs outrun linear surplus, and collapse five-hundredfold when a new kind of form unlocks. Its ceiling is external: complexity is logarithmic in environmental stability – which is algebra, since geometric build cost against linear surplus is logarithmic by inspection; the measured constant is 3.7 rungs per decade – so doubling complexity requires squaring how long the world holds still. We derive why partnerships are binary (error benefit saturates at two while seam cost keeps growing; break-even at 50.4% silent errors), why mutual gating between equals manufactures the rate hierarchy a system cannot give itself (13.87 bit-decades against 0.00), why a dead-end gut has an optimal feeding rate and a through-gut does not (a sign change, not a factor), and why every endotherm sits one to two bits of rate above ambient. We are explicit throughout about which results are measured, which are the algebra of a model we chose, and which are withdrawn.

1. The thesis and the ledger

Complexity is not a goal that systems pursue; it is what accumulates when a surplus is available and something can be built with it. The claim of this paper is narrower and more useful than "complexity increases": it is that the ORDER in which complexity arrives, and the HEIGHT at which it stops, are both fixed by an accounting that has nothing biological in it.

From the companion paper we take: a record system is a set of states with involutive moves; classes are the connected components of the move graph; R is the entropy of the class distribution; $H = R + H\_thermo$; R must never be read alone, because it is maximised when nothing moves. Growth supplies H_thermo and creates nothing; refusal converts H_thermo into R at a loss and terminates; a junction pays only when it blocks moves. Refusal requires an observer – something that distinguishes states – and the cheapest observer is one in which the measurement IS the decision.

A form is characterised by three numbers: a build cost B paid once, a maintenance cost m paid per unit time, and a return g. Over a lifetime T it pays iff $g > m + B/T$. Everything below follows from that inequality and from where its three terms can come from.

1a. What is not ours

The thesis that living systems are what a sustained energy flow builds is a century old: Lotka (1922) on maximum power, Odum and Pinkerton (1955), Morowitz (1968) on energy flow in biology, Schneider and Kay (1994) and Schneider and Sagan (2005) on gradient reduction. We claim none of it. The receptor optimum of Section 2 belongs to the family of discrimination ledgers opened by Hopfield (1974); that two is the number of copies a repair template needs is textbook; that the number of conservation laws of a reaction network is n_species - rank(S) is reaction-network theory (Schuster and Hofer 1991; Feinberg 2019). What is ours is narrower: the ORDER of construction and the CEILING, computed as consequences of one payback inequality inside the retention framework of the companion paper, together with the specific numbers of Sections 4 to 10, each labelled in Section 11 as measured, as algebra of the model, or as withdrawn.

2. Tier zero: geometry is free

At the origin there is no capital, so any form with $B > 0$ is unbuildable. We put this to a sweep: over 2000 random parameter sets spanning three orders of magnitude in build cost, maintenance and environmental richness, every single one produced zero rungs when every available form had a positive build cost. The ladder cannot start.

We found this by our own instrument failing – we had argued the constraint and then omitted a zero-cost form from the menu – which is how we recommend it be read: not as a modelling choice but as a constraint the model could not avoid. It is also not a finding: with no capital, $g > m + B/T$ fails for every $B > 0$ whatever the other terms, so 2000 of 2000 is what the inequality says, and the sweep is a check that the code implements the inequality. Section 11 files it as such.

The resolution is that the first tier costs nothing because it is not held. A cup that forms in a rock, a hole of a given diameter, a flow that admits particles below a certain size: these discriminate without maintaining a record, because they ARE the shape rather than a record of one. They are therefore the first observers, and they are free. The ordering is:

> TIER 0 – GEOMETRY. Free. Discriminates; holds no record; costs nothing to maintain. TIER 1 – FORM. A paid record, with a build cost and a maintenance cost, requiring surplus. TIER 2 – FUNCTION MODIFIES FORM. Self-reference, arriving last, and requiring a rate hierarchy that a single body cannot supply itself.

This also settles an ordering we could not otherwise resolve. Optimal chemical selectivity is b* = log2(c ln2 / mu) for a mistake cost c and a record decay rate mu, and above mu = c ln2 no maintained specificity pays at all: the optimal eater is indiscriminate. Geometric selection is adoptable at every mu, because it holds no record. Selectivity by shape necessarily precedes selectivity by chemistry.

3. Digestion before selection

Two functions differ structurally, and the difference forces their order. Digestion PRODUCES: its return scales with the environment – intake times yield. Selection SAVES: its return is bounded by the waste it prevents, hence by throughput.

Before anything digests, throughput is zero, so a selective form returns zero for any parameter values. This is not statistical, and it is not evidence either: it is a theorem of the two return functions, and the sweep below checks only that the code implements them (Section 11 files it so):

- Digestion adopted first in 100.0% of the 1512 sweeps that started.
- Selection adopted first in 0 of 2000. Exceptionless, as the asymmetry requires.
- Adoption order over a representative run: digestion -> selection -> structure, with the enabling form last because its return depends on there being something to enable. We state the full ordering more weakly than the first two facts: digestion-first and selection-never-first are sweep results over 2000 parameter sets, whereas the position of "structure" comes from individual trajectories and we have not swept it.

4. Surplus, not capability, drives the chain

A form that exactly repays its own maintenance is adoptable and contributes nothing to the next rung. Measured: with a positive surplus the ladder reaches fourteen rungs; with zero-surplus forms only – however capable – it reaches zero. The energy bonus is the mechanism, not a detail.

And the chain does not climb smoothly. Waits between rungs ran

1.05 0.42 0.46 0.32 0.44 0.67 1.08 1.82 3.16 5.59 10.04 18.25 | 0.03 0.06 0.11

shortening within a form type, lengthening as geometric build costs outrun linearly growing surplus, then collapsing five-hundredfold when a qualitatively new kind of form becomes affordable. Complexity compounds in bursts with stalls between them. We predicted monotone acceleration and were wrong; the burst structure is not built in, and follows from geometric costs against linear surplus plus the arrival of a new form type.

5. Waste, and the second opening

The binding constraint early is not energy but fouling: a one-aperture cup re-ingests its own output. With food value p, waste cost c, and re-ingestion saturating as r(i) = i/(i+K),

NET_one(i) = i p - c i^2/(i+K), NET_two(i) = i p.

NET_one has an interior maximum iff c > p, exactly at the predicted threshold, while NET_two is monotone. A sac-gut organism has an optimal feeding rate; a tube-gut organism does not. And the difference is a sign change, not a factor: the one-aperture form goes net negative for i >= 1. A dead-end gut that feeds too fast starves on its own waste.

A claim we withdraw. We argued that the second aperture makes "which end" a certificate – the through-gut as the first body axis – on the grounds that a second opening adds exactly one to the cycle rank of the flow graph. The topological half is a theorem: b1 = |E| - |V| + 1 goes 2 -> 3 -> 4 per aperture in every geometry we tried, and a single aperture is a bridge that no cycle passes through. The record half is not supported. In our dimer instances the extra coordinate carried no additional classes at four chamber sizes; whether a topological coordinate is REALISABLE depends on the geometry, not the cycle rank. Our instrument had four or five states and was too small to be decisive, so we report this as unsupported rather than refuted.

6. Two, and why

Mutual gating manufactures a hierarchy between equals. A record cannot shelter itself: free self-modification zeroes it. Shelter needs a slower gate, and a single body has no way to be its own slow partner. Two bodies do. Measured, at equal rates: 13.87 bit-decades under mutual gating against 0.00 under one-way gating and 0.00 uncoupled – and the advantage is largest exactly where the partners are most alike, vanishing when one is already a thousand times slower. The mechanism is deadlock, each partner holding the other still, with the assembly as a whole staying mobile.

And two is the optimum, not merely the minimum. The error benefit of redundancy SATURATES at two: a self-marking error – a break, an adduct, a mismatch recognised by shape – needs no vote to locate, and the intact copy is a repair template. A third copy adds correction only for *silent* errors. Meanwhile the seam cost KEEPS GROWING: at fixed substrate, going from two bodies to three loses 49.6% of the record, and the loss is monotone (19.87, 10.01, 5.02, 1.26 bit-decades at n = 2, 3, 4, 6). Putting both in one currency:

> Break-even at s* = 0.504: a third body pays only if more than half of all errors are silent.

In DNA nearly all damage self-marks, so two wins by a wide margin. Independently, holding units, substrate and contact budget fixed and varying only topology, record rises monotonically as contacts are concentrated – three seams 5.58, two seams 6.73, one seam 8.10 – and three bodies forced into a single seam converge on two bodies' efficiency. The cost is seams, not bodies.

7. The ceiling is the environment

An assembly must interface with an environment, and environments change. If the environment shifts on timescale T_env, no form's useful lifetime exceeds T_env, so payback becomes g > m + B/T_env. With build costs growing geometrically and surplus at best linearly, there is a computable last rung:

T_env	1e1	1e2	1e3	1e4	1e5	1e6	1e7	1e8	1e9	1e10
rungs	2.00	7.90	11.55	15.10	18.70	22.30	25.80	29.45	32.80	36.50

> Complexity is logarithmic in environmental stability: 3.7 rungs per decade, R^2 = 0.9966. To double complexity one must square the stability.

The slope tracks log2(10)/log2(beta) to within a constant 7.5% across build-cost growth factors from 1.5 to 6, so the ceiling is set by geometric build cost against linear surplus and by nothing else. That makes the logarithm algebra, not a discovery: a build cost B_0 beta^k against a surplus at most linear in T_env gives k_max = log_beta(T_env) + const by inspection, and the R^2 measures how well a fit to a logarithm fits a logarithm. What the sweep adds is the constant, the 7.5% band, and the sawtooth; Section 11 files the law itself under the model's own algebra. Losses are steps, not decay – the rung count is a sawtooth, since an environmental shift is a generator activating. Nothing forbids the twentieth rung; it needs a world that holds still a thousand times longer than the tenth did.

A control that mattered. Our first run of this experiment gave a clean-looking stall at 24 rungs and a plausible fit. It was wrong: a horizon parameter was acting as a finite T_env. The pre-registered check "at T_env = infinity the ladder must be unbounded" failed and voided the run. Without that line written down beforehand we would have published the artifact.

8. Branching

Complexity spreads like limbs rather than climbing like a ladder. Branches raise the ceiling, but only with descent – when a pruned limb is replaced by growth from a surviving limb rather than from zero. With descent, 128 branches add 4.02 rungs over one; without it, 1.93. The stability slope is reproduced independently at 3.73 rungs per decade.

> ~3.7 rungs per decade of stability against ~1.9 per decade of diversity:
> one decade of stability is worth roughly a hundredfold more branches.

Two runs of this experiment were voided by our own inconsistency – one measured an all-time maximum, the other a final snapshot, and they disagreed for that reason alone. Only the time-averaged sustained depth is reported.

9. The edge

A thermal gradient is a rate hierarchy laid out in space, and it is free. At a fixed observation horizon the two ends are two different deaths:

position	R	H_thermo	
hot	0	12	gas – melted
middle	6	6	alive
cold	12	0	crystal – frozen

R is maximal at the cold end, exactly where the system is most dead – the caveat of Section 1 appearing spatially, and a warning against scoring anything by R alone. We did exactly that and had to withdraw a measure that recommended being a crystal.

But the living condition holds at every position, at a *different horizon*: the alive window slides from T = 1e-5..1e-3 at the hot end to 1e3..1e5 at the cold. The gradient does not localise life to a place; it provides a place for every timescale. Wider spans are alive across more horizons – 1 of 13 at width 0.02, 11 of 13 at width 1.0 – so generalists buy timescale-breadth and specialists buy one timescale.

Vents pay twice. They are among the most stable environments available, which buys rungs at 3.7 per decade, and their edge supplies a rate hierarchy at no cost, which buys the observer. Neither requires the chemistry to be special.

10. Body temperature in bits

Rates are Arrhenius, so the framework's unit is bits of rate, not kelvin. One bit costs ln2 R T^2 / Ea = 10.4 K at Ea = 50 kJ/mol near 300 K.

	dT	bits of rate
bird (41C)	21 K	1.98
human (37C)	17 K	1.62
mouse (37C)	15 K	1.42

(Bits are computed exactly as (Ea / R ln2)(1/T_ambient - 1/T_body) at Ea = 50 kJ/mol, not as dT / 10.4 K; the linearised figure would read 1.64 for humans and 2.02 for birds.) | reptile basking | 10 K | 0.94 | | hibernating mammal | 5 K | 0.55 | | ectotherm, deep sea, thermophile | 0 | 0.00 |

The three endotherms tabulated sit between one and two bits above ambient, and hibernation is a deliberate drop to half a degree of freedom. We note the obvious limitation: three species is not a survey, the activation energy is taken as a single representative value when real Ea spans roughly 30-80 kJ/mol (which moves the bit count by a factor of about 2.7 either way, as the first column of the table shows), and "ambient" is a choice. What the table supports is an order of magnitude – one to two bits, not a tenth of a bit and not ten – and the observation that ectotherms sit at exactly zero, which requires no assumption about Ea at all. A model in which the first bit buys a degree of freedom and each further bit activates a generator that melts a bit of record, against a metabolic cost linear in dT, puts the optimum at 10 K – one degree of freedom. Humans sit at 1.62 bits.

The framework earns the order, not the value: why the margin is ~10 K rather than 1 or 100, from chemistry (Ea, T) and a one-generator floor. The residual above one bit is not ours to explain: the human figure is 1.62 bits at an ambient of 20 C, 1.13 at 25 C and 2.14 at 15 C, so the residual is a choice of ambient. (An earlier draft read it as the coupled-generator floor of the companion paper's Section 8a; that reading was post hoc on a free input and is withdrawn.)

10a. Why carbon, as algebra

Two facts about carbon are derivable in this framework's own terms, and a third would be numerology.

Tetravalence is the minimum that stores a record at a site. The record at a substituted centre is the number of orbits of substituent assignments under the site's ROTATION group – exactly a certificate in the sense of the companion paper, a value the moves cannot shift.

geometry	valence	orbits	record per site
trigonal planar (sp2)	3	1	0.00 bits
tetrahedral (sp3)	4	2	1.00 bit
trigonal bipyramidal	5	20	4.32 bits
octahedral	6	30	4.91 bits

Valence three admits no chirality and stores nothing; valence four stores exactly one bit (log2 of two orbits: orbit counting, no generator named). The chain algebra agrees: two bonds continue a backbone, leaving v-2 free, so four is the first valence that can polymerise AND carry information at every monomer.

Six-membered rings are angle algebra, and six strictly dominates. A strain-free planar n-ring requires 180(n-2)/n = theta, i.e. n = 360/(180-theta): sp2 at 120 degrees gives n = 6.000 exactly, sp3 at 109.4712 gives 5.104. Counting distinguishable substitution patterns modulo the dihedral group – the ring's own record capacity – against sp2 strain:

n	distances	isomers	record	sp2 strain
3	1	4	2.00 bits	3600
4	2	6	2.58	900
5	2	8	3.00	144
6	3	13	3.70	0
7	3	18	4.17	73
8	4	30	4.91	225

Six strictly dominates three, four and five on BOTH axes – less strain and more record – while seven and eight buy more record only by paying strain six does not pay. > Six is the first ring size at which distinguishable arrangements are available for free. > Below it one pays strain for less record; above it one pays strain for more. A hexagon is also the smallest ring with three distinct positional relationships (ortho, meta, para), which is the concrete content of "more distance and options".

What we decline. Carbon's atomic number six, the strain-free ring size six, and the constant C_2 = 6 of the framework that motivated this work are three different sixes: the ring size comes from 360/(180-120), the atomic number from nuclear charge. They coincide numerically and we have no derivation linking them. Asserting one would be numerology. What is defensible is that both results above are statements about move sets and their invariants, which is this framework's subject and not a coincidence of integers.

10b. Chemistry: reaction networks are move graphs

The bridge to chemistry is linear algebra, not analogy. A reaction network's stoichiometric matrix S gives reachable compositions x0 + image(S) – a coset – and its conservation laws are the left null space of S, functions constant along every reaction. Cosets and certificates: exactly the structure of the companion paper's Theorem 9. Hence

R = n_species - rank(S) = the number of independent conservation laws,

verified to integer precision on every network tried, and equal to the codimension of the orbit.

One reaction is one bit. Adding the steps of the hydrogen-combustion chain mechanism one at a time gives R = 6, 5, 4, 3, 2, 2 – four independent radical steps cost exactly one bit each, and the fifth (H + OH -> H2O) costs nothing because it is linearly dependent on the earlier four. The dependency was found by the computation, not supplied to it.

Catalysis destroys no record in this count, which we predicted, then wrongly retracted. We predicted that a catalyst adds reactions without adding a generator – pure rate, no cost – and an earlier draft reported the opposite, "R from 3 to 2" for Michaelis-Menten. That comparison was made against a baseline in which E and ES were present but inert, carrying two trivial conservation laws. Against the honest baseline, S -> P with no enzyme (R = 1), the catalysed network E + S <-> ES -> E + P has rank 2 and R = 2: catalysis adds the enzyme's own conservation law (E + ES) and removes none; the substrate pool's law S + P survives as S + ES + P. Catalysing one step of the combustion chain likewise raises R by exactly one (toy 5692, all seven predictions hit, hashed 04a652ef). What a catalyst does to a record is therefore entirely a matter of RATE: it moves a reaction across the observation threshold 1/T of Section 8b in the companion paper, adding a generator to the move set seen at that horizon without changing what the stoichiometry can ever conserve. An enzyme that accelerates racemisation destroys chirality at every horizon longer than its turnover; the stoichiometric record was never the chirality's protection. The sentence about enzyme specificity that an earlier draft drew from the rank drop is withdrawn with it; what survives is the rate statement of the genome paragraph below.

What a metabolism costs. Over a pool of eight species with all interconversions possible but frozen, each enzyme costs one bit until the reactions become dependent: R runs 8, 7, 6, 5, 4, 3, 2, 1 and then stays at 1. A complete metabolism destroys all of a pool's record but the total mass.

How a genome survives one. Not by chemical difference – the same atoms, and the scrambling reactions are allowed. A record dies when a generator that moves it enters the move set, and at horizon T the move set is the reactions faster than 1/T, so the genome survives because no enzyme catalyses the moves that would scramble it: its sequence sits below the metabolic threshold. With measured figures – an uncatalysed phosphodiester half-life of order 30 million years against a metabolic step of order a second, and a nuclease acceleration of 10^17 – the protection margin is 49.8 bits of rate and the enzyme supplies 56.5, a surplus of about seven. A nuclease is more than sufficient to melt what it acts on, so its specificity and spatial control are not refinements but the mechanism of preservation.

11. What is measured, what is model, what is withdrawn

Measured, and we stand behind: the 14 rungs of the surplus ladder (the 0 of the capability ladder is the bootstrap inequality again and belongs below); burst structure; the ceiling's constant, its 7.5% band across beta and its sawtooth; branching with descent; the seam cost (49.6% from two to three) and the concentration series; the 13.87 bit-decades of mutual gating (the 0.00 is the companion paper's Section 8b theorem, that a record which can unshelter itself has R = 0); the sliding alive-window's numbers.

Computed on published inputs with a chosen ambient, neither measured nor ours: the endotherm bit table (Section 10).

The algebra of a model we chose, verified numerically but not independent of its construction: the bootstrap (2000/2000 – g > m + B/T fails at zero capital for every B > 0, so the sweep is a code check); digestion-first (100% / 0 of 2000 – a saving function returns zero at zero throughput, likewise a code check); the logarithmic ceiling and its slope law (geometric build cost against linear surplus is logarithmic by inspection, and R^2 = 0.9966 measures a fit to a logarithm fitting a logarithm); the receptor optimum b* = log2(c ln2/mu) and its corollaries; the fouling functional form AND its sign change (an interior maximum of ip - c i^2/(i + K) exists iff c > p, by differentiation); the two spatial deaths (hot = R at 0, cold = H_thermo at 0, which is R(T) under an Arrhenius gradient by construction); the zero side of the capability ladder; the break-even s* = 0.504, which depends on our measured seam cost being the right price; the stoichiometric integer checks of Section 10b (rank-nullity). An earlier draft listed several of these under "measured"; a sweep over a deterministic implication is a check on the code, not evidence for the model.

Withdrawn in the course of this work: that the second aperture creates a body axis (topology yes, record unsupported); that a universal balance R/H characterises living systems – twice, first because a uniform parameter sweep is not an ensemble of living systems, and then again when a narrower re-test still turned out to be reporting its own grid, since the abelian steady state has the closed form R/H = 1 - f + 1/g which reproduces every measured value to 0.023; a thermal trade-off model whose code was linear where its prose was saturating; and the claim that complexity accelerates monotonically. What survives of the balance is a one-sided bound: sustaining systems sit above R/H = 0.6 in every case tested, never near the 1/2 we predicted and never at either death. The recurring lesson, which cost us the same error twice in one day: *a tight distribution over a swept parameter is a property of the sweep until shown otherwise.*

Instrument failures caught by controls: a grid coarse-graining blind to KAM tori; a canonicalisation that made "the pair" mean "the two lexicographically smallest"; a horizon masquerading as an environmental timescale; an inconsistent statistic between two runs; a mis-scaled gradient. Five, all caught, two of them by pre-registered kill criteria that fired as written. Predictions for every experiment were hashed before the corresponding run; the digests are retained.

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